14 Volt Power System (Mod S.O.O. 6272, 6281 and 6282) – General data

1. General

There are three 14 volt power outlets configuration are available in the cabin to provide power to peripherals: cell phone, laptops, etc.

2. Description

A. STA 332 14V Convenience Outlet – (Mod S.O.O. 6272)

There is a single 14 volt power outlet may be installed on the forward right hand side of STA 332 bulkhead in the cabin. A 14 volt convenience outlet on the forward, right hand side of STA 332 bulkhead. The outlet has a maximum load 7.5 amps. The 28 Volt DC to 14 Volt DC power converter is located on the aft avionics racks, mounted underside of the top shelf below the CVR.

28 Volt DC power is provided from the right bus, through a 5 amp circuit (CB18) located on the power distribution circuit box. 14 Volt DC power is provided to the outlet and is protected by a 7.5 amp STA 332 circuit breaker on the aft converter circuit breaker panel located near the power converter.

B. Aft Cabin 14V Convenience Outlet – (Mod S.O.O. 6281)

There are three 14 volt power outlets may be installed between STA 220 and 332 bulkhead in the cabin. Each outlet has maximum load 7.5 amps. The 28 Volt DC to 14 Volt DC power converter is located on the aft avionic racks, mounted underside of the top shelf below the CVR.

28 Volt DC power is provided from the right bus, through a 15 amp circuit (CB18) located on the power distribution circuit box. 14 Volt DC power is provided to each of the three outlets and each outlet is protected by a 7.5 amp STA 223, STA 258, STA 332 circuit breakers located on the aft converter circuit breaker panel located near the power converter.

C. Fwd Cabin 14V Convenience Outlet – (Mod S.O.O. 6282)

There are three 14 volt power outlets that may be installed between STA 111 and 186 in the cabin. Each outlet has maximum load 7.5 amps. The 28 Volt DC to 14 Volt DC power converter is located on the forward avionics compartment, mounted in under the middle shelf. In addition to powering the three cabin outlets the power converter also powers the existing two 14 volt outlets located in the cockpit, on the pilot and co-pilot's side of the center pedestal, refer to 25-11-00, 14 Volt Power Converter System – General Data.

The 28 Volt DC power is provided from the right bus through a 15 amp PWR CONVR 28V INPUT (M6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data). 14 Volt DC power is provided to the existing cockpit outlets through a 7.5 amps PWR CONVR 14V OUTPUT (N6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data), and to the cabin outlets through a 7.5 amp STA 128 14 VDC, STA 176L 14 VDC, STA 176R VDC circuit breaker located on the forward converter circuit breaker panel located near the power converter.

3. Adjustment/Test

A. Function Test of 14 Volt Power Converter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

3) Plug an accessory into each outlet, as required, and ensure the accessory is being powered by the outlet.

4) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal/Installation

A. Removal of 14 Volt Power Converter for Mod S.O.O. 6282

1) Ensure the aircraft electrical system is OFF.

2) Open the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel, and the CB18 circuit breakers located on the power distribution circuit box and fit safety clips.

3) Gain access to the forward avionics shelves through the forward baggage compartment.

4) Remove the middle and bottom covers attached to bulkhead 44.00 as required.

5) Disconnect the electrical connector.

6) While supporting the 14V Power Converter, remove the four mounting screws.

7) Remove unit from shelf.

Note Stow connectors as required.

B. Removal of 14 Volt Power Converter for Mod S.O.O. 6272 and 6281

1) Ensure the aircraft electrical system is OFF.

2) Open the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Gain access to the 14 Volt Power Converter, which is accomplished through the aft baggage compartment into the rear compartment.

4) Disconnect the electrical connector.

5) While supporting the 14V Power Converter, remove the four mounting screws.

6) Remove unit from shelf.

Note Stow connectors as required.

C. Installation of 14 Volt Power Converter for Mod S.O.O. 6282

1) Ensure the aircraft electrical system is OFF.

2) Ensure the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel.

3) Set unit in position and install four mounting screws.

4) Reconnect electrical connector.

5) Install the middle and bottom covers attached to bulkhead 44.00. Ensure the covers are mounted securely.

6) Remove the safety clips and close the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel.

7) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

8) Do a function test as per Para 3.A.

D. Installation of 14 Volt Power Converter for Mod S.O.O. 6272 and 6281

1) Ensure the aircraft electrical system is OFF.

2) Ensure the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Set unit in position and install four mounting screws.

4) Reconnect electrical connector.

5) Remove the safety clips and close the CB18 circuit breakers located on the power distribution circuit box.

6) Do a function test as per Para 3.A.

E. Removal of 14V Power Outlets for Mod S.O.O. 6282

1) Ensure the aircraft electrical system is OFF.

2) Open the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel and fit safety clips.

3) Remove the countersunk screws from the outlet mounting plate.

4) Remove outlet assembly from the interior paneling.

5) Gently separate the outlet electrical disconnect. Note which disconnect came off the negative terminal and which disconnect came off the positive terminal, indicated on back side of plug.

6) Unthread the metallic locknut from the back of the outlet.

7) Remove outlet by pulling it out the front of the mounting hole.

Note Stow the connectors as required.

F. Removal of 14V Power Outlets for Mod S.O.O. 6272 and 6281

1) Ensure the aircraft electrical system is OFF.

2) Open the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Remove the countersunk screws from the outlet mounting plate.

4) Remove outlet assembly from the interior paneling.

5) Gently separate the outlet electrical disconnect. Note which disconnect came off the negative terminal and which disconnect came off the positive terminal, indicated on back side of plug.

6) Unthread the metallic locknut from the back of the outlet.

7) Remove outlet by pulling it out from the front of the mounting hole.

Note Stow the connectors as required.

G. Installation of 14V Power Outlet for Mod S.O.O. 6282

1) Ensure the aircraft electrical system is OFF.

2) Ensure the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel, and the CB18 circuit breakers located on the power distribution circuit box and fit safety clips.

3) Install the outlet in mounting panel from the front of the panel.

4) Thread the metallic locknut onto the back of the outlet and tighten. Ensure the outlet is secured in the panel.

5) Install the electrical connectors onto the appropriate terminals on the rear of the outlet.

6) Install outlet assembly from the interior paneling.

7) Install the countersunk screws from the outlet mounting plate.

8) Remove the safety clips and close the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel, and the CB18 circuit breakers located on the power distribution circuit box.

9) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

H. Installation of 14V Power Outlet for Mod S.O.O. 6272 and 6281

1) Ensure the aircraft electrical system is OFF.

2) Ensure the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Install the outlet in mounting panel from the front of the panel.

4) Thread the metallic locknut onto the back of the outlet and tighten. Ensure the outlet is secured in the panel.

5) Install the electrical connectors onto the appropriate terminals on the rear of the outlet.

6) Install the countersunk screws from the outlet mounting plate.

7) Remove the safety clips and close the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

8) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

5. Troubleshooting

A. Ensure associated system of the circuit breakers are closed.

B. Visually inspect the outlet connectors for damaged wiring or contacts. Ensure electrical connectors on outlet are fully seated and secure.

C. If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in PSM 1-64-2W.

D. Measure 14V output voltage at PWR CONV.